The Grounded Research Object

Metric-informed design of a machine-native scientific record whose unit computes its own credibility

Carla Ostmann

desciencemodel.com · Positron · carla@positron.vc — Working paper, July 2026

Abstract. Research assessment runs on citation-derived indicators despite two decades of documented dysfunction: citations reward citable outputs and are structurally blind to negative results, replications, refutations, and reuse. I ask whether indicators derived from the *structure* of a machine-readable research record can measure — and let funders reward — what citations cannot. To test this I drafted 64 candidate contribution indicators over a structured corpus — an existing agent-native record format (ARA; Liu et al., 2026) adopted as the departure point for this work — and selected them through blind adversarial tournaments. The central result is negative and clarifying: **metrics computed over a record's own structure measure the fidelity of the record, not the quality of the science**; genuine signal appeared only at the claim level, anchored to external ground truth. Auditing every blocked indicator against the data shape that blocked it shows most failures are format problems, not feasibility problems — the record does not lack the knowledge, it lacks the shape. From that audit I derive the Grounded Research Object (GRO): a record format that sorts every quality signal into three rigor tiers — deterministic, reliable-anchored, and reproducible-judged — kept physically separate so no self-certified number is mistaken for a checked one, and rolled into a funder-facing credibility dossier that never collapses to a single number. GRO is a specification, adversarially stress-tested but empirically unvalidated; I state its limits plainly and argue the two binding constraints — validation and adoption — are not protocol problems.

1. Introduction

The paper is the unit of scientific output, and the citation is its score. Once that holds, Goodhart's law does the rest: when a measure becomes a target it ceases to be a good measure. Careers, grants, and tenure denominated in citations push researchers toward citable work — reviews, positive results, broad confident narratives — and away from the labor everyone agrees is undervalued: negative results, replications, refutations, shared data, methods reused downstream. The reform literature (DORA; the Leiden Manifesto; *The Metric Tide*; CoARA) is mature but largely *subtractive* — use fewer numbers, more narrative — with few constructive proposals for indicators that measure the neglected contributions, and with Goodhart usually asserted against any new metric rather than tested.

Meanwhile the record itself is changing. AI agents increasingly read, reproduce, and extend research, and machine-readable structured records — typed claims, evidence links, exploration traces — are emerging as a publication layer. For the first time there are artifact surfaces over which non-citation indicators could actually be computed. This paper reports what happens when you try, and what the attempt reveals about the record we would need. The governing thesis is blunt: **reward what citations punish**, and defend against gaming not with one clever number but with a diversity of independent signals and a selection process in which no metric is graded by its own author.

2. The metrics program

I organised the work around seven directions of good-science signal: (1) reproducibility and specification completeness; (2) claim and evidence integrity; (3) process transparency and *negative knowledge* — dead-end density, failure-knowledge preservation, negative-result share; (4) novelty and dependency structure, including a corrective-science score weighting refutations above imports; (5) a cross-library *claim graph* whose edges are evidential (corroboration, contradiction, replication depth, reuse) rather than social — a direct alternative to the citation network; (6) real-world grounding against trial registries, chemical databases, and datasets; and (7) representation quality as a trust weight.

These became 64 concrete candidate indicators drafted across every surface a structured record exposes, pruned and merged to 23 survivors and a top ten. Selection ran as two blind, adversarial tournaments so a metric could not be graded by whoever proposed it. In round one, for each of eleven artifact surfaces, four clean-room proposers (no sight of any existing metric) proposed indicators; a meta-science judge picked the best two; both were revised against critique; one won. In round two, each top-ten survivor ran a four-to-two-to-one tournament over three refinement cycles. Proposers were drawn from two independent model families (GPT-5.5/codex and Claude Sonnet 5) and the meta-science judge was a separate model (Claude Fable), so no metric was graded by the model that proposed it. The tournaments were run over a real corpus — roughly 140 Alzheimer's-domain papers, about 60 compiled into the ARA format (Liu et al., 2026) — so the indicators were not merely designed but *run*, and measured.

3. Result: structure measures fidelity, not quality

The headline is negative, and it is the most useful output of the exercise:

> *Metrics computed over a record's own structure measure the fidelity of the record, not the quality of the science.*

A well-compiled record of bad science and a well-compiled record of good science were indistinguishable to every structural metric in the pool. Zero of six paper-level rankers survived the validity check. Genuine signal appeared only at the claim level, and only when it reached *outside* the record: claims checked against registered clinical-trial endpoints, priority claims checked against prior literature, and one compiler-introduced polarity inversion caught. A direct measurement of extractive fidelity against full text ran at roughly 92% faithful, with the failures concentrated in dense multi-cohort comparison papers where a transposed table flips a result.

The lesson is not that structural metrics are impossible. It is sharper: the signal worth rewarding lives outside the record's own walls, and the record's *shape* determines whether you can reach it.

4. The affordance gap

For every metric I wanted but could not build, I asked why. Almost never because the information was missing — because it was stored in a shape only a language model could re-extract: a number hand-retyped as prose in four places instead of one typed value, citations as author-year strings instead of resolvable identifiers, honest absence indistinguishable from lazy omission. The record did not lack the knowledge; it lacked the shape. Every blocked metric sorts into one of three classes, and each tells the format what to do.

Class	Why the metric is blocked	What the format must do
Format-recoverable	the fact is present but as prose an LLM must re-read	emit it typed → the metric becomes a deterministic join
Anchor-dependent	the fact points outside the record and nothing resolves the pointer	guarantee resolvable external IDs so registries and indices can be joined
Irreducibly semantic	no format change makes it computable	a calibrated judge, forever — the format only feeds it cleaner, pinned inputs

The first two classes are format problems masquerading as feasibility problems. Only the third is a true ceiling. This taxonomy is the bridge from a metrics result to a substrate design.

5. The Grounded Research Object

GRO gives the record a shape. Every load-bearing fact exists once as a canonical typed record, addressed by an identifier; prose binds back to it. The narrative paper survives as a *compiled view*, not the source of truth. Everything a record must carry is sorted into three rigor tiers, kept in physically separate files so a self-certified number can never be dressed as a checked one.

Tier	What it holds	How far it can be trusted
DETERMINISTIC	typed quantities, claim logical form, typed cross-layer graph, genre manifest	exact and reproducible — a structural join
RELIABLE-ANCHORED	references, registrations, accessions, datasets	reliable given a pinned resolver; failures quarantined, not faked
REPRODUCIBLE-JUDGED	novelty, significance, entailment quality, assumption realism	a calibrated judge, forever; the format only feeds it cleaner inputs

A decisive structural invariant separates two write-surfaces: the compiler writes only the skeleton facts it can ground, while independent resolver, extraction, and judge passes write everything that *certifies* those facts, into append-only files a validator forbids the compiler from touching. The tiers roll up into a funder-facing credibility dossier that never collapses to one number — a program officer can gate on the deterministic floor, rank on the judged tier, read the rest as risk, and audit any score back to its source *and its rigor class*. Honest absence — `not_specified`, `not_applicable`, `unresolved` — is a first-class value scored equal to an honest answer, so a typed field never creates pressure to fabricate.

The design was not hand-authored. Each of twelve affordance gaps ran its own four-proposer design tournament; the twelve winning designs were merged into one specification and put through an adversarial red-team pass that forced every over-claim down to a stated limitation. GRO is a clean-sheet design informed by, but not inheriting from, the format it critiques.

6. Where existing efforts fall short

Three current efforts each measure a different axis, none calibrated against the others, and none covering the reward-what-citations-punish set. **Citations** measure social attention. A structural **rigor seal** (the closest kind of automated review) measures internal *integrity* — coherence, falsifiability, grounding — as a single grade, but never asks whether the work is novel or matters, and is calibrated to nothing external. External benchmarks such as the Metascience Novelty Indicators Challenge measure *novelty*, calibrated to expert ratings, but only novelty, and they read whole papers rather than a structured record. Integrity, novelty, and fidelity are three orthogonal axes; the contributions citations punish fall between them. Closing that gap is a substrate problem first — most of the ideal indicator set cannot be computed until the record carries the right shapes.

7. Limitations

GRO is a specification, adversarially hardened but **empirically unvalidated**. The honest limits are as important as the design, and several are drawn directly from the two research programs this work feeds.

7.1 It has not been shown to discriminate

The open experiment — the one that would tell us whether any of this is real — has not been run: take the affordance-derived metrics, run them over existing literature (starting from the testbed corpus, adding a contrasting second domain), and validate against external ground truth such as trial-registry concordance, retraction and correction records, and prior-literature novelty. Until then, "GRO makes the metric computable" is not "the metric separates good science from bad." The negative result of §3 is a warning that this must be shown, not assumed. Running that discrimination test — extending these metrics over a labelled corpus and calibrating them against funder judgements of what to reward — is the subject of the author's doctoral programme at SPRU (University of Sussex). **Update (2026-07): a first discrimination test has now been run**, on the breakthrough axis, over 66 recent and 72 historical (2004–2010) Alzheimer's papers (see repository `experiment/breakthrough/`). The result is sobering and is reported in full: the one metric that carries signal is the L8 contribution-typing (`max(peak,cwmean)`). Against a same-model LLM expert panel it reached $\rho \approx 0.58$ (held-out); against an independent model family only $\rho \approx 0.34$ (roughly a third was shared-method bias — the lean flips when the metric is built by the other model, Steiger $p=0.003$); and against a *real-world, LLM-free ground truth* — mature citation-disruption of the historical papers over the following 15–20 years — it showed **no reliable predictive power** (ρ near zero, sign-unstable). The prior-art `overlap`/`sota_anchor` axis, the delta ledger, and genre added no transferable signal. So on the evidence so far the metric flags *LLM-perceived contribution depth*, which is not shown to be *field-reshaping impact*. This sharpens rather than removes §7.1: the discrimination test is now begun, its first verdict is largely negative, and the honest next step (a full-text, multi-domain historical corpus, calibrated against real outcomes) is the subject of the author's doctoral programme at SPRU (University of Sussex).

7.2 Importance is judged, not computed — and does not scale

Standardisation buys *fidelity*-measurability cheaply and deterministically. It does not buy *importance*-measurability: novelty, validity, and contribution significance remain in the judged tier, needing a calibrated judge and human-labelled ground truth. That tier does not scale to a full corpus — deterministic and anchored signals can run at population scale, but calibrated judgment runs only on a triage set plus an audit sample. The corpus-scale promise is therefore honestly "deterministic and anchored credibility at scale; calibrated-judged credibility at triage-and-sample scale," which means the format does not, on its own, surface the un-shortlisted gem.

7.3 A record that computes its own credibility relocates trust; it does not eliminate it

The anti-fabrication guarantees rest on an independence between automated checkers that is, throughout, *asserted rather than measured*: genuinely decorrelated checking is earned at exactly one non-LLM point in the design; everywhere else "independent" means "a different model," and two models can share a blind spot. So credibility is moved from trusting the author's prose to trusting a calibration set and an assumed independence — real progress, because the trust is now auditable, but not the free lunch the phrase suggests. Calibration sets and probe sets are themselves institutional trust anchors that can be captured, and cross-discipline comparison remains a modelling choice, withheld rather than faked where a discipline cannot be resolved.

7.4 Adoption is the wall, and it is owned by incentives

Every prior attempt at a better medium — nanopublications, research objects, and my own earlier work — failed not on architecture but on incentives: no reason for a researcher to assemble structured objects when grants and jobs are denominated in papers. A better specification does not move that wall. What has changed is the cost side: agents now sit inside the research workflow and the trace they generate is the raw material, so capture can be largely automated — necessary, nowhere near sufficient. And the record is extensible by design: agent-native substrates such as ARA (Liu et al., 2026) and Claude Science take new data shapes through adapters and plugins scoped to the target metric, so the format can grow to carry a chosen indicator's inputs without a rewrite. The binding questions — which field's researchers are the real early adopters, and whether funder co-design can shift incentives at meaningful scale — are not protocol problems, and this work does not solve them. They are, deliberately, the subject of the applied research programme this specification supports: the empirical Goodhart test of these indicators under adversarial optimisation, and the study of what evaluation arrangements funders and evaluators would regard as legitimate.

8. Next steps

Two moves, both closing the gap between "designed" and "shown to work." First, **run the discrimination experiment** (§7.1): compile existing literature into the format and validate the metrics against external ground truth, testing at the same time whether the three affordance classes hold under adversarial pressure. Second, **build the metrics with funders**. Funders are where measurable becomes rewarded; the metrics over the record should be shaped by what funders actually want to reward, and the affordance model tells them what the record must carry for a chosen indicator to be valid rather than gameable. Concretely: name the undervalued contributions to reward; supply the human-labelled ground truth (as the Novelty Challenge did for one axis); and fund the shapes, resolvers, and calibration sets as shared public goods. Non-traditional funders are already piloting new publishing practices as funding conditions; the larger innovation agencies could sharpen their own metric efforts with an affordance-aware view of the record.

9. Conclusion

Chasing good metrics far enough hands you the specification for the record itself. The metrics are the entry point — the lever, because nothing changes until what counts changes, and the diagnostic, because trying to compute them reveals what the substrate must be. GRO is that substrate, designed backwards from a metrics failure rather than forwards from taste, and honest about the two things still missing: proof that the metrics discriminate, and an institution willing to make the good ones count. Neither is a protocol problem, which is precisely why the work now turns to experiment and to funders.

Appendix A. The data shape — sidecar reference (as emitted)

The complete set of typed sidecars the compiler emits per record today, in the style of an API schema: each with its id space, rigor tier, field definitions, and a real example (drawn from a compiled single-cell Alzheimer's paper). Prose in `PAPER.md` binds back to these ids. This is the *emitted* subset of the full SPEC; layers the spec defines but the compiler does not yet materialize (the L4 cross-object graph, derived claim-graph AST, L7 registry joins, P1–P4 report objects) are listed in SPEC §7a.

quantities.yaml **Q## · Tier A (deterministic) · L1 — canonical typed numbers**

Every load-bearing number authored *in this paper*, typed once and addressed by id; prose and all other layers cite the `Q##`, never re-type the value.

field	type	meaning
`id`	string	canonical id (Q01, Q02, ...)
`value`	number	the number itself
`unit`	string	unit / measure type (e.g. log2FC, FDR, percent)
`comparator`	enum	none | vs_baseline | vs_arm — what the number is measured against
`ci_low / ci_high`	number|not_specified	confidence-interval bounds if reported
`role`	enum	result | input | parameter
`source_ref`	string	location in the source (e.g. "paper.pdf p6")
`quote`	string	verbatim sentence the value was lifted from (grounding)
`claim_refs`	[C##]	claims this number supports

```
quantities:
  - id: Q01
    value: 1.75
    unit: log2FC
    comparator: none
    role: result
    source_ref: "paper.pdf p6"
    quote: "...increase in the proportion of Ex5 neurons in high
            compared to low pathology cases in BA9 (log2FC = 1.75)"
    claim_refs: [C01]
```

external_quantities.yaml **XQ## · Tier B (anchored) · L1' — off-paper prior-work numbers**

Prior-work numbers used as comparison baselines. A *separate id space* from `Q##` (they carry someone else's `fetch_timestamp`, and are exempt from this paper's numeral-coverage audit).

field	type	meaning
`id`	string	XQ01, XQ02, ...
`value`	number	the prior-work number
`unit`	string	unit
`description`	string	what it is and which prior work it comes from
`source_anchor.external_id`	DOI/id	the resolvable source of the number
`source_anchor.fetch_timestamp`	datetime	when it was fetched
`baseline_verification`	enum	source_verified | self_reported | unverified

```
external_quantities:
  - id: XQ02
    value: 3.03
    unit: percent
    description: "Ex5 proportion of excitatory cells in the Green et al.
                  2024 prefrontal reference (19,360 / 637,968 cells)"
    source_anchor:
      external_id: "10.1038/s41586-024-07871-6"
      fetch_timestamp: 2026-07-07T00:00:00Z
    baseline_verification: source_verified
```

claims_typed.yaml **C## · Tier A (deterministic) · L2 — claims in logical form**

Each load-bearing claim, typed and given a first-order-logic line so contradictions and scope are machine-checkable. Binds to the quantities, entities and evidence that support it.

field	type	meaning
`id`	string	C01, C02, …
`claim_type`	enum	comparison | existence | causal | association | …
`polarity`	enum	positive | negative | null
`logical_form`	string	FOL rendering of the claim
`population_n`	int|not_specified	sample size if stated
`population_scope`	enum	cohort | subgroup | invitro | model | …
`quantity_refs`	[Q##]	numbers the claim rests on
`concept_refs`	[EN-*]	entities involved
`proof_refs`	[E##]	experiments/evidence
`depends_on`	[C##]	claim dependencies

```
claims:
  - id: C01
    claim_type: comparison
    polarity: positive
    logical_form: "increases_with(Ex5_proportion, pathology, BA9)
       AND significant(GLMM_FDR=0.008, BA9) AND trend(p=0.06, BA17)"
    population_scope: cohort
    quantity_refs: [Q01, Q02, Q03, Q04]
    concept_refs: [EN-population-ex5, EN-concept-resilience_neuronal]
    proof_refs: [E03]
    depends_on: [C02, C10]
```

entities.yaml **EN-* · Tier A (deterministic) · L3 — entity spine**

One record per entity mechanically referenced by another layer (genes, cell types, methods, concepts), with a stable id and optional ontology xref so records interoperate.

field	type	meaning
`id`	string	EN-<kind>-<slug>
`term`	string	human-readable name
`kind`	enum	concept | measure | method | dataset | organism | …
`xref`	ontology id|not_specified	external ontology anchor if resolved

```
entities:
  - id: EN-population-ex5
    term: "Ex5 (resilient L4 IT excitatory subtype)"
    kind: concept
    xref: not_specified
  - id: EN-marker-kcnip4
    term: "KCNIP4 (KChIP4 / CALP)"
    kind: measure
    xref: not_specified
```

genre.yaml — · Tier A (declared + checked) · L5 — genre contract

The record's declared type plus which content slots that type *expects*; absence of an expected slot becomes a declared, checkable fact rather than a silent omission.

field	type	meaning
`paper_type`	string	controlled genre label
`expected_slots`	[string]	slots this genre should carry
`present_slots`	[string]	slots actually present
`absent_declared`	[string]	expected-but-absent, declared honestly

```
paper_type: molecular_neuropathology_single_cell_omics
expected_slots: [claims, experiments, evidence, dataset]
present_slots:  [claims, experiments, evidence, dataset]
absent_declared: []
```

refs.yaml R## · Tier B (anchored) · L6 — resolved reference spine

One canonical record per cited work, resolved to an external id. Other layers cite the `R##`; unresolvable references are flagged, not hidden.

field	type	meaning
`id`	string	R01, R02, ...
`raw`	string	the reference as printed
`external_id`	DOI/id	resolved canonical id
`resolvable`	bool	whether it resolved to a real external record

```
refs:
  - id: R03
    raw: "Mathys et al., 2023 — single-cell atlas of cognition/
          resilience to AD [ref 8]"
    external_id: 10.1016/j.cell.2023.08.039
    resolvable: true
```

contributions.yaml CT## · Tier C (judged) → L8 · L8 — typed contributions (the breakthrough-metric input)

What the paper claims to contribute, *double-typed*: how the authors frame it vs. the compiler's own call from a closed novelty taxonomy, with a rationale. An anti-puffery lock voids any contribution not realized in a real claim. **This is the field the breakthrough metric reads.**

field	type	meaning
`id`	string	CT01, ...
`author_framed_type`	string	the authors' framing of the contribution
`compiler_assessed_type.primary`	enum	new_paradigm | new_method | new_finding | refutation | synthesis | incremental_improvement | replication
`compiler_assessed_type.confidence`	0–1	compiler's confidence in the type
`typing_rationale`	string	why this type
`typing_divergence`	enum	none | adjacent | conflicting (author vs compiler)
`adjudication`	enum	not_triggered | <blind-second-rater outcome>
`realized_in`	[C##]	claims that realize it (empty ⇒ voided)

```
contributions:
  - id: CT02
    author_framed_type: key_resilience_factor
    compiler_assessed_type: { primary: new_finding, confidence: 0.75 }
    typing_rationale: "Nominates KCNIP4 as a molecular resilience
        factor upregulated in resilient Ex5 L4 neurons; new gene-level
        finding grounded in prior KChIP4 biology (imported, not invented)."
    typing_divergence: none
    adjudication: not_triggered
    realized_in: [C04, C05, C06, C09]
```

delta_ledger.yaml **D## · Tier B (anchored) · L8 — comparative deltas vs prior work**

Each quantified comparison of this paper's number against an off-paper baseline. `claimed_value` is a `Q##`, `baseline_value` an `XQ##`; deltas are compiler-computed, never author-entered. Honesty states (qualitative / unresolved) are first-class, not blanks.

field	type	meaning
`id`	string	D01, ...
`delta_status`	enum	quantified | qualitative_only | claimed_unresolved | not_claimed
`claimed_value`	Q##	this paper's number
`baseline_value`	XQ##|not_specified	the prior-work number
`absolute_delta` / `relative_delta`	number|not_specified	compiler-computed
`baseline_verification`	enum	source_verified | self_reported
`note`	string	what the comparison is and why the status

```
deltas:
  - id: D02
    delta_status: quantified
    claimed_value: Q16          # 34.42% (this paper, BA17)
    baseline_value: XQ04        # 0.33% (SEA-AD DLPFC atlas)
    absolute_delta: 34.09
    relative_delta: 103.303
    baseline_verification: source_verified
    note: "Ex5 spatial enrichment vs prefrontal atlas — establishes
           BA17-specialization of the resilient L4 IT subtype.
```

sota_anchor.yaml — · **Tier B (resolver-produced) · L8 — prior-art precedence neighborhood**

A resolver-produced neighborhood of the closest *preceding* works, each with an `overlap` ∈ [0,1] (how much of this paper's contribution the prior work already covers). Frozen at an externally-timestamped `precedence_date`. Read as **novelty = 1 − overlap** or **convergence = + overlap** (see §7.1).

field	type	meaning
`precedence_date / cutoff_date`	date	the frozen prior-art horizon
`provenance`	enum	openalex_resolved | reference_resolved | compiler_estimated
`neighborhood[].ref`	R##	a prior work in the neighborhood
`neighborhood[].overlap`	0–1	fraction of this paper's contribution it already covers
`neighborhood[].contemporaneous_uncited`	bool	near-neighbor the paper failed to cite
`overlap_jaccard_second_resolver`	0–1	independent second-pass corroboration
`retrospective_resolution`	object	later supersession signal + gap_days

```
precedence_date: 2026-01-01
provenance: openalex_resolved
neighborhood:
  - {ref: R03, overlap: 0.35, contemporaneous_uncited: false}  # Mathys
       2023 — framework-level overlap only (research program, not this
       paper's specific L4-IT/KCNIP4 finding)
  - {ref: R05, overlap: 0.30, contemporaneous_uncited: false}  # same lab,
       vulnerability direction, not resilience
overlap_jaccard_second_resolver: 0.22
```

temporal.yaml — · **Tier B (anchored) · temporal spine**

Publication and fetch timestamps that pin every anchored/forward-looking signal to a time axis (required for precedence and any longitudinal metric).

field	type	meaning
`pub_date`	date	publication date
`doi`	string	canonical id
`fetch_timestamp`	datetime	when external anchoring was performed
`note`	string	provenance note

```
pub_date: 2026-01-01
doi: "10.1038/s41467-026-68920-4"
fetch_timestamp: 2026-07-07T00:00:00Z
note: "pub_date seeds the corpus temporal spine"
```

Appendix. Method and provenance

The metric indicators were selected by two blind adversarial tournaments (round one per artifact surface, ~88 subagents; round two per metric, four-to-two-to-one over three refine cycles). The GRO specification was produced by twelve per-gap design tournaments (four independent proposers each — two on one model family, two on another — a critical meta-science judge selecting and critiquing the best two, a refine round) whose twelve winning designs were merged and then subjected to an adversarial critique pass that forced every over-claim to an explicit limitation or a design change; design-tournament run identifiers `wf_f0bc615b-a88` and tail `wf_c4cbff37-887`. All artifacts — the seven directions, the 64→23→10 indicator ledger, the negative-result report, the affordance-gap taxonomy, the full GRO specification, both tournaments with their winning proposals and judgements, and the compute code including the cross-library claim graph — are collected in the accompanying repository (`github.com/carlaost/gro`). This document is a digest of a longer compiled reference; where it summarises, the repository carries the full text.

This work adopts the ARA format (Liu et al., 2026) as the substrate it critiques and stress-tests; GRO is the substrate-design output of that critique, not a re-implementation. Positioning, evidence pointers, and the honest-limitations detail are documented in the repository and in the author's research program map.